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DATA – 6950

Capstone II – Professor Weijie Pang

Summer 2026

GreenBookAI: An AI-Assisted Travel Recommendation System

Motivation / Problem

- Travelers must compare information across multiple websites and services.
- Recommendations often prioritize popularity over individual preferences.
- Important factors such as weather, budget, safety, and attractions must be evaluated separately.
- Many recommendation systems provide little transparency into why destinations are suggested.

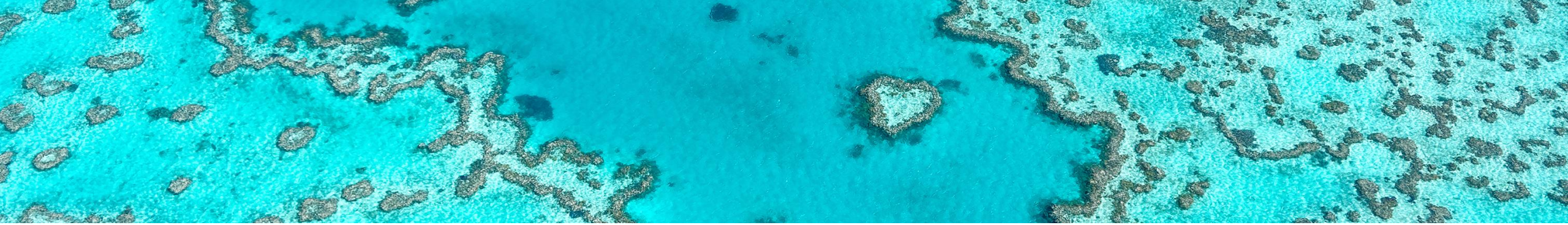
- Accepting travel requests in natural language.
- Combining multiple travel factors into a single recommendation process.
- Enriching recommendations with live travel information.
- Providing AI-generated explanations to help users understand each recommendation.



Project Objectives

- Interpret travel requests written in natural language.
- Extract traveler preferences such as budget, destination type, travel duration, and travel season.
- Identify destinations that best match the user's preferences through a multi-stage recommendation pipeline.
- Enrich recommendations with live travel information, including weather and travel advisories.
- Generate clear, AI-assisted explanations that help users understand why destinations were recommended.





System Architecture

Backend Framework

- FastAPI exposes REST endpoints and coordinates application workflows.

Modular Services

- Recommendation logic, live data retrieval, and AI explanation are separated into independent services.

Response Generation

- Results from the recommendation pipeline are combined with AI-generated explanations before being returned to the user.

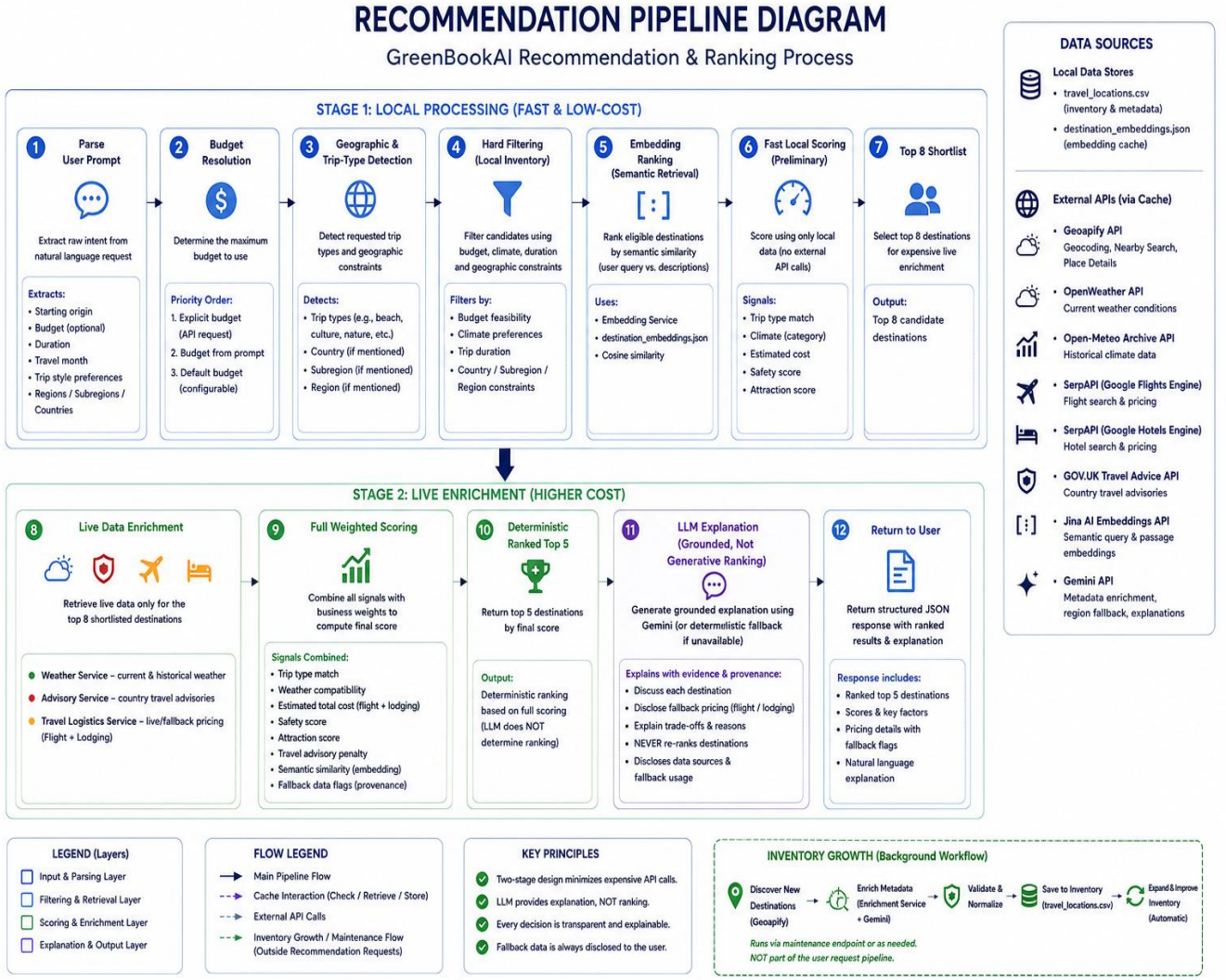


Stage 1 – Candidate Selection

- Interpret the user's travel request.
- Filter destinations using extracted preferences.
- Rank candidates through semantic retrieval and local scoring.

Stage 2 – Recommendation Refinement

- Enrich top candidates with live travel information.
- Compute final recommendation scores.
- Generate AI-assisted explanations for the final recommendations.



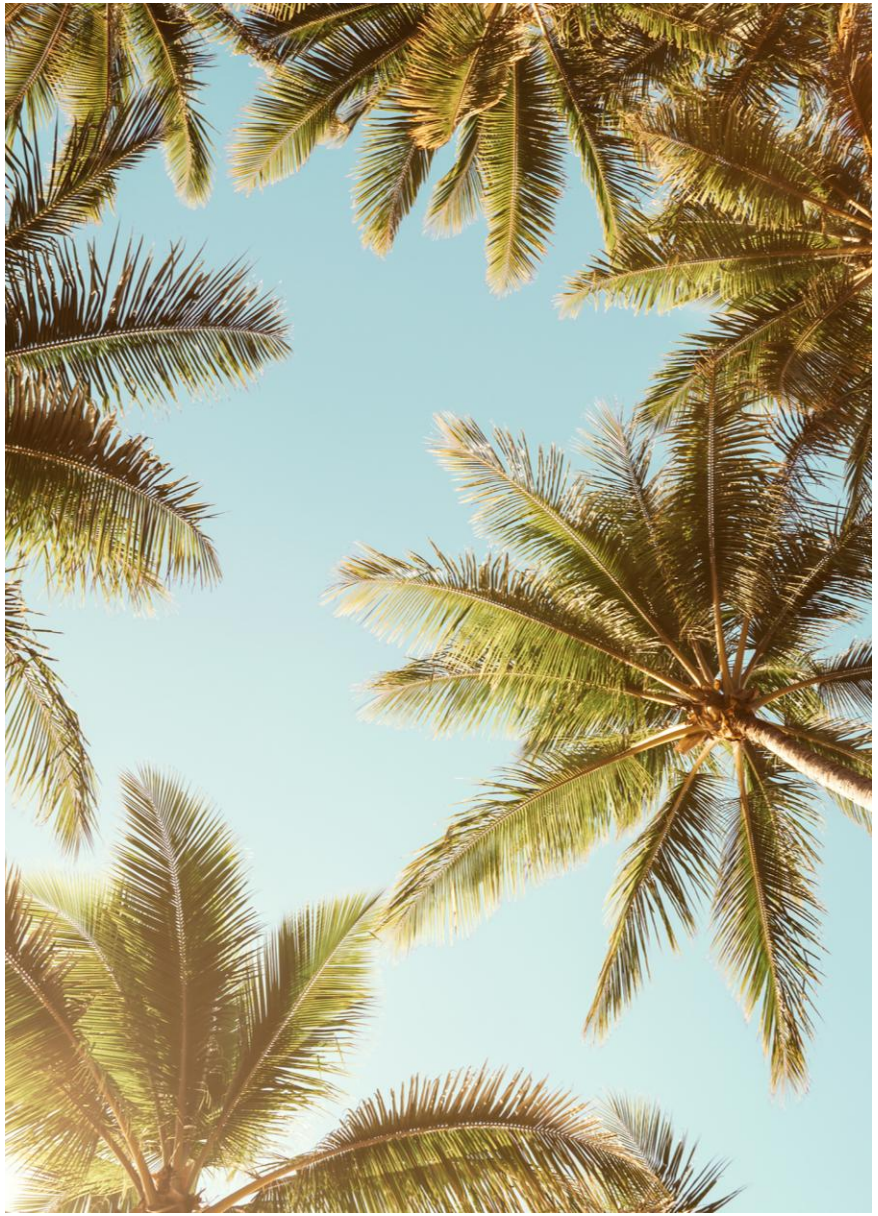
Semantic Retrieval Using Embeddings

```
0.04868903, 0.09211376, -0.06939466, 0.08633237,
```

Process

- **Receive the user's travel request as a structured JSON payload** containing preferences such as destination, budget, and travel prompt.
- **Apply hard filters** to remove destinations that do not satisfy explicit requirements (such as trip type, region, or subregion when specified).
- **Convert the user request and each remaining eligible destination into numerical embeddings (vectors).**
- **Calculate cosine similarity** between the request embedding and each destination embedding to measure semantic relevance.

embedding_score = max(similarity, 0) × 60





Hybrid Weighted Ranking

- **Combine the embedding score (cosine similarity × 60) with the preliminary local score to calculate a combined preliminary ranking score.**
- **Apply weighted rewards and penalties based on factors such as trip type, region, subregion, climate, budget, safety, and attractions.**
- **Rank all eligible destinations by their combined preliminary scores and select the Top 8 candidates.**
- **Enrich the Top 8 with live weather, seasonal climate, estimated pricing, and travel advisory data before calculating the final recommendation scores.**

Combined Preliminary Score= max(Cosine Similarity,0)×60+T+G+C+B+S+A+P

Symbol	Factor
T	**Trip Type Match**
G	**Geographic Match** (Region/Subregion)
C	**Climate Preference Match**
B	**Budget Compatibility**
S	**Safety Score**
A	**Attraction Score**
P	**Safety Preference Bonus** *(applied only when the user requests safer destinations)*

Example Scoring Rules
Trip Type: +40 match / −40 mismatch
Subregion: +45 match / −45 mismatch
Region: +35 match / −35 mismatch
Climate: +20 match / −5 mismatch (warm/cold)
Budget: +25 within budget / +5 within 30% / −30 over budget



Understanding the User's Request

"I want a warm beach vacation in March for about a week with a budget under $2,000."

User Request	Extracted Preference
Warm	Climate Preference
Beach vacation	Trip Type
March	Travel Season
About a week	Trip Duration
Under $2,000	Budget



Live Data Enrichment

Why Live Data?

- Only the strongest candidate destinations are enriched with live data.
- Improves recommendation accuracy using current travel conditions.
- Reduces unnecessary external API requests.

External Services

- **OpenWeather** – Retrieves current weather conditions.
- **Open-Meteo Archive** – Provides historical climate data.
- **Geoapify** – Supplies geographic and location information.
- **Travel Advisory Service** – Evaluates destination safety advisories.
- **Travel Logistics Service** – Estimates travel costs and logistics by combining hotel and flight data.

Explainable AI

Role of Gemini

- Generates human-readable explanations for recommended destinations.
- Summarizes why each destination matches the traveler's preferences.
- Improves transparency and user understanding.

Recommendation Process

- Destination rankings are determined **before** Gemini is called.
- Gemini receives the ranked recommendations as input.
- The recommendation engine remains deterministic and explainable.

Benefits

- Improves user trust through clear explanations.
- Separates AI-generated explanations from recommendation logic.
- Makes recommendations easier to interpret.



Technical Implementation

Backend and Validation

- **Python** powers the application and recommendation logic.
- **FastAPI** exposes the backend endpoints and coordinates the workflow.
- **Pydantic** validates incoming requests and structures application data.
- **Uvicorn** runs the FastAPI application.

Data and Integration

- **Pandas** loads and processes the local destination inventory.
- **Requests** connects GreenBookAI to external travel-data services.
- **python-dotenv** loads API keys and configuration from environment variables.
- **cachetools** supports caching where repeated data access would be inefficient.

AI and Explanation

- **Semantic embeddings and cosine similarity** support candidate retrieval.
- **Google Gen AI SDK** connects the application to Gemini.
- **Gemini** generates natural-language explanations after ranking is complete.



Current Progress

Completed

- FastAPI backend and modular service architecture
- Two-stage recommendation pipeline
- Natural language preference extraction
- Semantic retrieval and destination scoring
- Live API integration (weather, climate, location, advisories)

Gemini-generated recommendation explanations

In Progress

- Expanding the destination dataset
- Refining recommendation accuracy and scoring weights
- Improving generated information
- End-to-end testing and performance evaluation



Remaining Work

Recommendation Quality

- Reduce reliance on fallback information by increasing live data coverage.
- Continue refining the recommendation scoring model for greater accuracy.

Richer Destination Data

- Expand the destination dataset with additional attractions and points of interest.
- Include common activities, local experiences, and seasonal events for each destination.

Scalability & Performance

- Improve caching and data refresh strategies.
- Continue expanding destination coverage and overall recommendation quality.

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Demonstration

"I want a warm beach vacation in March for one week with a budget under $2,000."





Conclusion

- GreenBookAI transforms natural language travel requests into personalized destination recommendations.
- A two-stage recommendation pipeline combines semantic retrieval, deterministic scoring, and live travel data.
- Gemini enhances the user experience by generating clear, explainable recommendations.

Future Vision

Advanced Recommendation Methodology

- Incorporate machine learning to learn recommendation weights from historical user interactions instead of using fixed scoring.
- Evaluate hybrid recommendation approaches that combine rule-based scoring with collaborative or content-based filtering.
- Investigate retrieval-augmented generation (RAG) techniques to improve explanation quality using verified travel knowledge.

Richer Data Integration

- Integrate additional travel data sources, including attractions, local events, transportation, and tourism datasets.
- Reduce reliance on fallback information by expanding real-time and historical data coverage.
- Replace static destination attributes with dynamically refreshed travel information.





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Q&A